

Grade 8
Unit 14
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

1. A ten-ounce container of mixed nuts includes peanuts, walnuts, cashews, and almonds and has a total of 320 pieces. Complete the following probability distribution so that it is valid.

	Almonds	Cashews	Peanuts	Walnuts	Total
Frequency	80	80	120	40	320
Probability	$\frac{1}{4}$, 0.25 or 25%	25%	$\frac{3}{8}$	0.125	1

2. Derrick has a playlist of 50 of his most listened to songs. Complete the table by using the probability to determine how many songs of each music genre he has in his playlist.

	Country	Pop	Rap	R&B	Rock
Frequency	5	10	10	5	20
Probability	$\frac{1}{10}$	0.2	20%	0.1	40%

Sheila works at the local clothing store. The store has just received a shipment of new inventory. The shipping list says that 800 different clothing items are in the shipment. Sheila determined the following:

$P(\text{dresses}) = 0.1$
 $P(\text{shorts}) = 0.15$
 $P(\text{accessories}) = 0.375$

$P(\text{jeans}) = 0.25$
 $P(\text{graphic tees}) =$

3. Sheila forgot to determine the probability of selecting a graphic tee from the new inventory. Explain how she can determine the missing probability with the information she already calculated.

All of the probabilities should add up to 1. She can add the probabilities she already has and subtract that from 1.

4. What is the probability of selecting a graphic tee?

$$1 - (0.1 + 0.15 + 0.375 + 0.25) = 0.125$$

5. How many dresses are in the new inventory?

$$0.1 \times 800 = 80 \text{ dresses}$$

6. How many jeans are in the new inventory?

$$0.25 \times 800 = 200 \text{ jeans}$$

7. How many shorts are in the new inventory?

$$0.15 \times 800 = 120 \text{ shorts}$$

8. How many graphic tees are in the new inventory?

$$0.125 \times 800 = 100 \text{ graphic tees}$$

9. How many accessories are in the new inventory?

$$0.375 \times 800 = 300 \text{ accessories}$$

10. Construct a probability distribution to check your work.

	dresses	shorts	accessories	jeans	graphic tees	total
Frequency	80	120	300	200	100	800
Probability	0.1	0.15	0.375	0.25	0.125	1